

Homework #4

AMS 507, Fall 2025

Due October 2

1. Let Y be the number of successes throughout n independent repetitions of a random experiment having the probability of success $p = 1/4$. Determine the smallest value of n so that $P(Y \geq 1) \geq 0.7$.
2. Let $X \sim \text{Bin}(2, p)$ and $Y \sim \text{Bin}(4, p)$. If $P(X \geq 1) = 5/9$, find $P(Y \geq 1)$.
3. If $x = a$ is the unique mode of $X \sim \text{Bin}(n, p)$, show that

$$(n+1)p - 1 < a < (n+1)p.$$

4. If $X \sim \text{Poisson}(\lambda)$ such that $P(X = 1) = P(X = 2)$, find $P(X = 3)$.
5. Let the number of geese in a certain pond have a Poisson distribution. We want the probability that at least two geese are in the pond at a certain time to be greater than 0.99. Find the smallest value that the mean of the distribution can take.
6.
 - (a) Calculate the factorial moment $E[X(X-1)]$ for the binomial distribution.
 - (b) Calculate the factorial moment $E[X(X-1)]$ for the Poisson distribution.
 - (c) Use the result of part (a) to calculate the variance of the binomial distribution.
 - (d) Use the result of part (b) to calculate the variance of the Poisson distribution.
7. A chicken sexer makes 1 error in every 500 examinations when he detects the sexes of chicks.
 - (a) Find the exact probability that he makes no more than 2 errors in 1,000 examinations.
 - (b) Using the Poisson distribution, find an approximate probability that he makes no more than 2 errors in 1,000 examinations.
8. In a classroom of 30 students, 3 of the students wear wrist watches.
 - (a) If 14 students are selected with replacement, what is the probability that exactly 2 of them wear wrist watches?
 - (b) If 14 students are selected without replacement, what is the probability that exactly 2 of them wear wrist watches?

9. An apartment seeker in Manhattan estimates that 10% of the available apartments in her price range are in acceptable condition. Furthermore, she has time to look at only one apartment every weekend.
 - (a) How many weeks should this person expect to spend for apartment hunting?
 - (b) What is the probability that she will find an acceptable apartment in the first two weekends that she looks?
 - (c) What is the probability that it will take more than 4 months (17 weeks) to find an acceptable apartment?
10. In a class of 100 students, 25 students have hard cover and 75 students have paperback textbooks for the course. If you randomly choose 10 students in this class, find the probability that 2 of them have hard cover texts in the following ways:
 - (a) the exact probability
 - (b) approximate probability using a binomial distribution
11. Suppose that $P(X = 0) = 1 - P(X = 1)$. If $E(X) = 3\text{Var}(X)$, find $P(X = 0)$.
12. An urn contains n balls, numbered 1 through n . If m balls are randomly withdrawn in sequence, each time replacing the ball selected previously, find $P(X = k)$, $k = 1, \dots, n$, where X is the maximum of the m chosen numbers.